



Preparation and performance analysis of carbon-based composite superabsorbent polymers

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Received: 16 August 2025 / Accepted: 27 November 2025 / Published online: 2 December 2025
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Abstract

Global water scarcity and soil desertification underscore the critical need for efficient, cost-effective water-retentive materials in sustainable agriculture. To address this, graphene oxide (GO), carbon nanotubes (CNT), and rice husk biochar (RBC) were incorporated into an acrylic acid/2-acrylamido-2-methylpropanesulfonic acid/sodium alginate copolymer network, fabricating three carbon-composite superabsorbent polymers (GO-SAP, CNT-SAP, RBC-SAP). Systematic evaluation revealed that carbon materials dispersed uniformly within the polymer matrix via hydrogen bonding and van der Waals forces, with surface oxygen-containing groups significantly enhancing hydrophilicity. XRD and SEM analyses confirmed amorphous porous structures, with GO-SAP and RBC-SAP exhibiting superior pore distribution. Maximum swelling ratios occurred at loadings of 0.08 g (GO), 0.12 g (CNT), and 0.12 g (RBC). At equivalent loadings, GO-SAP demonstrated the highest absorption capacity, attributable to GO's strong hydrophilicity, 2D lamellar structure, and exceptional dispersibility. All composites maintained high swelling ratios between pH 5–10 and 20–50 °C but showed sensitivity to multivalent ions. Swelling kinetics followed a pseudo-second-order model ($R^2 > 0.99$), indicating chemically controlled absorption governed by hydrophilic groups. Remarkably, the composites retained water for >4.5 h at 50 °C and preserved >97% water retention after centrifugation (12,000 rpm). Cyclic stability exceeded 80% of initial swelling capacity after 6 absorption-desorption cycles. This work provides a robust foundation for developing high-performance, sustainable superabsorbents.

Keywords Superabsorbent polymers · Carbon materials · Swelling · Resuse

Introduction

Escalating global water scarcity and soil desertification underscore the urgent need to develop efficient and eco-friendly water-retentive materials for sustainable agriculture and ecological restoration [1]. Superabsorbent polymers (SAP), characterized by their unique three-dimensional polymeric networks, exhibit immense potential in

arid-region agriculture and soil amendment due to their exceptional water absorption/retention capacities, well-developed porosity, and favorable biocompatibility [2].

However, conventional synthetic SAP often suffer from poor acid/alkali and salt tolerance, while natural polymer-based SAP face limitations in stability [3]. To overcome the performance constraints of single-component SAP, material hybridization strategies have emerged as a key research focus [4]. In this context, carbon nanomaterials have garnered significant attention for enhancing SAP properties due to their unique physicochemical characteristics, with considerable progress already demonstrated [5]. For instance, carbon nanotubes (CNT) have been incorporated into SAP matrices, leveraging their high specific surface area and exceptional mechanical strength to improve the mechanical integrity and structural stability of the composite hydrogels [6]. Biochar (BC), with its hierarchical pore structure and abundant surface functional groups, has been utilized to enhance the water-holding capacity and nutrient release

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properties of SAP composites [7]. Similarly, graphene oxide (GO), featuring a two-dimensional lamellar structure rich in oxygen-containing functional groups, serves as an ideal cross-linking site for constructing more robust and stable polymeric networks [8]. The schematic chemical structures of CNT, GO, and BC are depicted in Fig. 1. These studies collectively confirm that incorporating carbon materials into SAP systems can significantly enhance overall performance through interfacial interactions and synergistic effects, presenting a promising avenue for developing high-performance composite SAP [9].

Despite this considerable promise and the significant advances achieved in enhancing specific SAP properties based on the distinct structural attributes and surface chemistry of individual carbon materials, a systematic comparative analysis of their differential impacts remains notably scarce. Existing research predominantly focuses on employing a single carbon material to improve one or a few specific properties, such as mechanical strength or absorption rate [10]. However, a comprehensive comparative investigation into the differential enhancement effects and the underlying mechanisms exerted by structurally diverse carbon materials—specifically, quasi-zero-dimensional BC, one-dimensional CNT, and two-dimensional GO—on critical SAP performance metrics (e.g., absorption capacity, absorption rate, water retention, swelling strength, and cyclic stability) within an identical polymer matrix under comparable conditions is still lacking. This knowledge gap hinders the precise evaluation of which carbon material is optimal for specific performance requirements and consequently constrains the rational design and material selection based on intrinsic structural advantages. While our research group has prior expertise in this field, the core originality of this work lies in its pioneering effort to systematically compare the differential modifying effects of three structurally distinct carbon materials on an identical SAP matrix within a unified platform. This approach aims to elucidate universal structure-property relationships between carbon material characteristics and SAP performance, thereby transcending

the conventional research paradigm focused solely on single-material optimization or isolated property enhancement.

Therefore, this study constructs three composite SAP systems by individually hybridizing an identical polymer matrix with CNT, BC, and GO. We systematically investigate and compare the influence of these carbon materials on the resin network structure, water absorption/retention kinetics, and overall performance within the same composite system. By establishing clear structure-property relationships between the characteristics of the carbon materials and the performance of the composite SAP, this work provides a direct theoretical and experimental basis for selecting the most suitable carbon material to develop high-performance composite SAP tailored to specific application scenarios, such as water-saving agriculture where long-term cyclic stability is paramount.

We fully acknowledge the potential environmental risks associated with SAP based on acrylic acid monomers, alongside cross-linkers like N, N-methylenebisacrylamide and initiators such as ammonium persulfate. Although trace amounts of acrylic acid monomer are considered biodegradable, concerns regarding their ecotoxicity persist [11]. Building on our previous research, the synthesis conditions in this study have been optimized to minimize and remove unreacted monomers and small-molecule by-products effectively, thereby mitigating their potential adverse environmental impacts. Future work will include a detailed ecotoxicity assessment of the final products, which is a crucial step toward their practical application.

In summary, this study aims, through a controlled experimental approach, to clarify the differential enhancement mechanisms of carbon nanomaterials with different dimensionalities (CNT, BC, GO) on SAP performance. Its core novelty resides in providing a scientific guideline for the selection of carbon materials based on specific functional requirements, thereby advancing the precise design and application of efficient and environmentally friendly water-retentive materials in sustainable agriculture and ecological restoration.

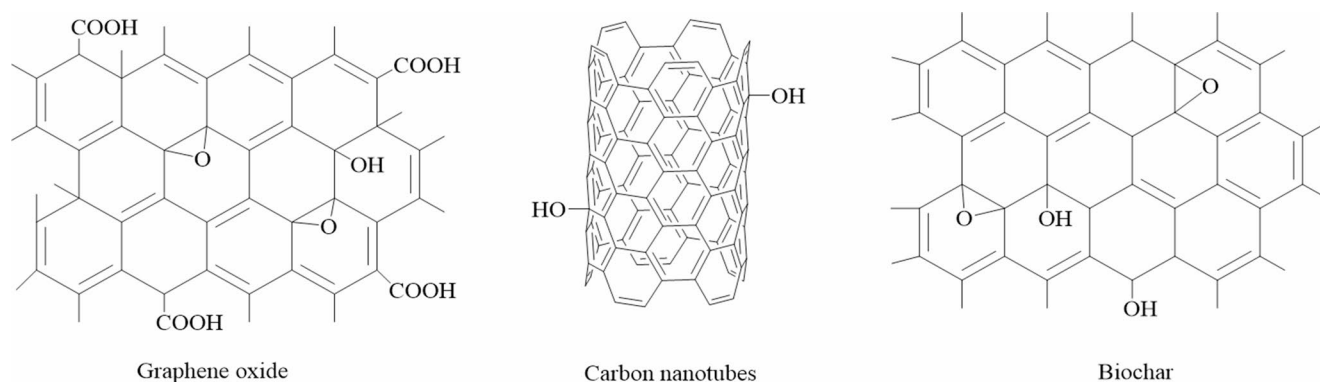


Fig. 1 Schematic diagram of the general structure of GO, CNT and BC

Materials and methods

Materials

Acrylic acid (AA, 99%), 2-acrylamido-2-methyl-1-propane sulfonic acid (AMPS, 98%), sodium alginate (SA, CP), N, N-methylenebisacrylamide (NMBA, 99%), ammonium persulfate (APS, 98%), sodium hydroxide (96%), graphene oxide (GO, 99%), carbon nanotubes (CNT, 95%), sodium chloride (99%), calcium chloride (98%), hydrochloric acid (36.5%) were purchased from Sinopharm Chemical Reagent Co., Ltd, China. Rice husk biochar (RBC) was purchased from Henan Lize Environmental Protection Technology Co., Ltd, China. Deionized water, tap water, lake water (Qingfeng Lake, Dongying, China, electrical conductivity: 1096 $\mu\text{S}/\text{cm}$), and river water (Guangli River, Dongying, China, electrical conductivity: 4740 $\mu\text{S}/\text{cm}$) were used for the experiments.

Preparation of GO-SAP, CNT-SAP and RBC-SAP

Taking one of the polymerization processes as an example, 1.5 g of SA was fully dissolved in 35.0 g of deionized water to obtain Solution (1). Next, 6.4 g of NaOH was dissolved in 20.0 g of deionized water. Subsequently, 14.4 g of AA and 4.0 g of AMPS were added to this solution and thoroughly dissolved to yield Solution (2). Solutions 1 and 2 were then mixed uniformly. After that, 0.1 g of GO, 0.05 g of APS, and 0.04 g of NMBA were added to the mixture. The resulting solution was subjected to ultrasonic dispersion until homogeneous. The solution was then placed in an oven at 70 °C and allowed to react for 4 h. After the reaction, the gel was removed, cut into small pieces, dried at 105 °C, pulverized, and passed through a 40–60 mesh standard sieve. Finally, GO-SAP was obtained.

To compare the effects of CNT and RBC on the properties of the SAP, the polymerization conditions for RBC-SAP and CNT-SAP were kept identical to those used for GO-SAP.

Determination of sap's properties

Swelling property

A measured mass of the SAP sample was immersed in a sufficient volume of test solution. After complete swelling, the gel was separated from excess deionized water by filtration through a nylon mesh bag. The mass of the swollen gel was recorded. The water absorption capacity Q (g/g) was calculated using Eq. (1) [12]:

$$Q = \frac{M_2 - M_1}{M_1} \quad (1)$$

Where M_1 and M_2 represent the masses of the SAP before and after water absorption, respectively.

Water retention performance

The water-saturated hydrogel was weighed and placed in a petri dish or centrifuge tube. After exposure to specified conditions for predetermined time intervals, the remaining hydrogel was retrieved and weighed. For the tests at ambient conditions, the saturated hydrogel was kept in the open laboratory environment to simulate natural evaporation. For the tests at elevated temperatures, the samples were placed in a controlled oven. The water retention rate R (%) was calculated using Eq. (2) [13]:

$$R = \frac{M_t}{M_0} \times 100\% \quad (2)$$

Where M_t is the mass of the hydrogel at time t , M_0 is the mass of the initial saturated hydrogel.

Repetitive water absorption performance

A measured mass of dry SAP was first immersed in 0.01 mol/L NaCl solution or lake water until fully saturated. The swollen hydrogel was then filtered and weighed. Subsequently, the water-saturated hydrogel was dried in an oven at 105 °C until constant weight was achieved. This absorption-drying cycle was repeated 15 times, with the water absorption capacity calculated after each cycle using Eq. (1).

Swelling kinetics

A pre-weighed portion of SAP was placed into a nylon mesh bag (40 mm $\times$ 50 mm, 0.076 mm pore size). After sealing the bag and recording its weight, it was immersed in an excess of deionized water. At 10-minute intervals, the bag was removed, drained of excess water, left to stand for 5 min, and then weighed. The water absorption capacity of the SAP at different time intervals was calculated using Eq. (1).

Structural characterization

Fourier transform infrared (FTIR) spectroscopy was performed using a Nicolet iN10 spectrometer (Thermo Fisher Scientific, USA). Morphological analysis was observed under an FEI QUANTA FEG450 scanning electron microscope (FEI Company, USA). Crystal structure characterization was conducted via X-ray diffraction (XRD) using a PANalytical X'Pert Pro diffractometer (Malvern Panalytical,

UK) with Cu-K α radiation ($\lambda = 1.5406 \text{ \AA}$) operated at 40 kV and 100 mA. XRD scans were recorded from 5° to 90° (2θ) at a scan rate of $5^\circ/\text{min}$.

Results and discussion

Structural characterization

The FTIR spectrum presented in Fig. 2(a) reveals that GO, CNT, and RBC exhibit characteristic absorption peaks at approximately 3400 cm^{-1} (corresponding to O-H stretching), around 1600 cm^{-1} (attributed to C = C stretching), and near 1080 cm^{-1} (indicative of C-O stretching). These peaks confirm the presence of hydrophilic oxygen-containing functional groups in all three materials [14]. Notably, the additional peak at 1725 cm^{-1} (C = O stretching) observed in the GO spectrum suggests a more abundant content of oxygen-containing functional groups in GO compared to the other two materials. Moreover, the broader hydroxyl peaks observed for both GO and RBC, as opposed to CNT, imply stronger hydrophilicity in GO and BC.

Figure 2(b) demonstrates that the FTIR spectra of pure SAP and the three composite SAP are highly similar. Major characteristic peaks are observed at 3300 cm^{-1} (O-H stretching), 2930 cm^{-1} (CH_2/CH_3 stretching), 1552 cm^{-1} and 1403 cm^{-1} (associated with $-\text{COO}^-$ groups), 1180 cm^{-1} (S = O stretching), and 1043 cm^{-1} (C-O-C stretching). These peaks successfully validate the formation of an AA-AMPS-SA polymer network [15]. It is worth noting that, although no new characteristic peaks emerge in the composite SAP, there are shifts in the positions of the absorption peaks corresponding to certain functional groups. This indicates that the carbon materials are integrated into the resin matrix primarily through physical interactions such as multiple hydrogen bonds and van der Waals forces, rather than through the formation of new chemical bonds.

As illustrated in Figs. 2(c) and 2(d), GO, CNT, and RBC exhibit primary diffraction peaks at 2θ values of 11.2° , 26.1° , and 26.3° , respectively. The XRD patterns of the three composite superabsorbent resins all display a broad diffraction peak around $2\theta = 20.8^\circ$, which clearly indicates that the polymer structure is amorphous. Moreover, the diffraction peaks corresponding to the three carbon materials disappear in their respective composite SAP, suggesting that these carbon materials are uniformly dispersed within the polymer matrix.

SEM results (Fig. 2e) reveal that the surfaces of the three composite SAP exhibit irregular and non-uniform morphological features, accompanied by pronounced porous structures. The presence of these pores implies that the SAP possess a larger surface area, providing more space for the

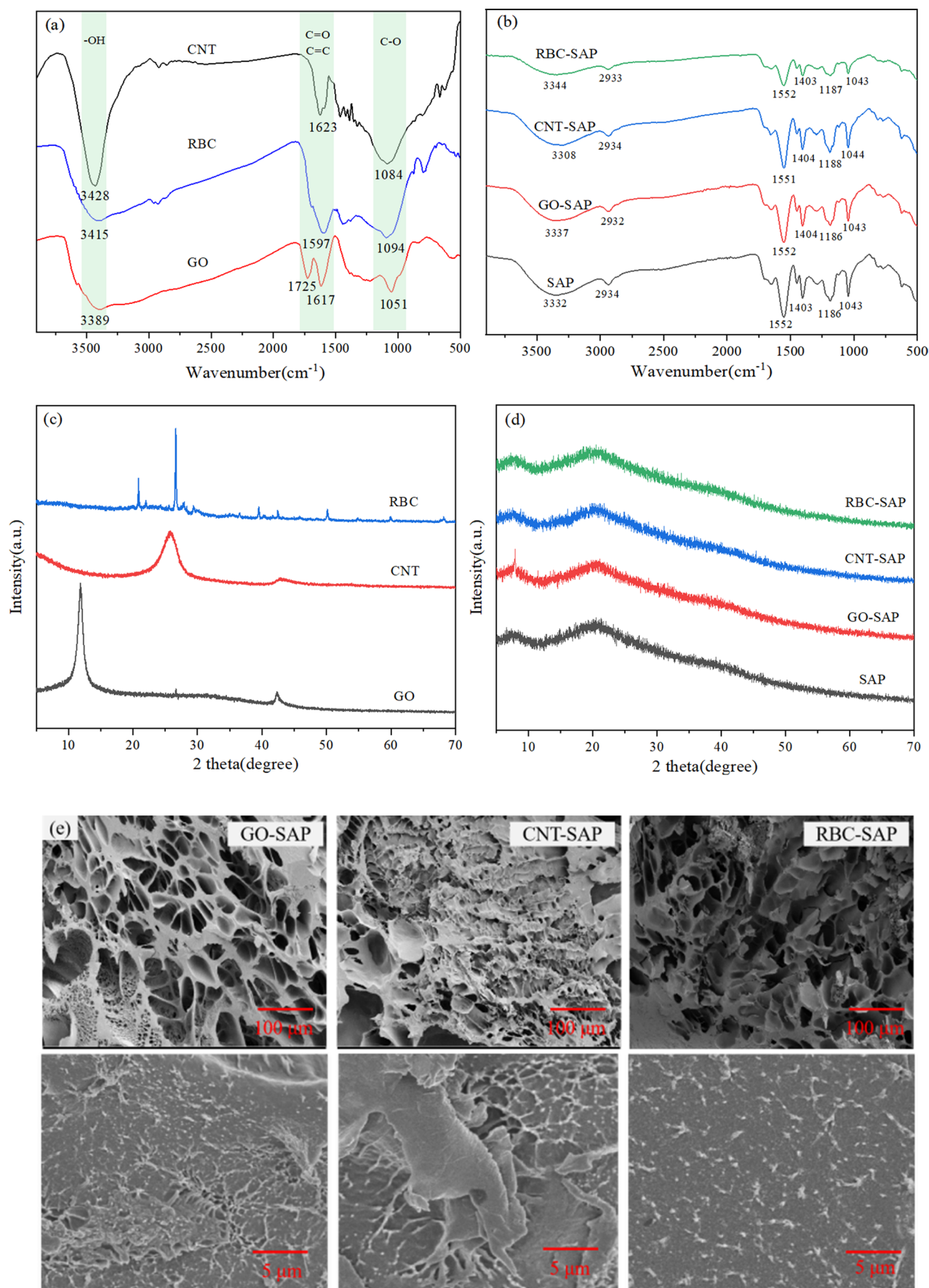
Fig. 2 FTIR spectra of (a) GO, CNT, RBC (b) pure SAP, GO-SAP, CNT-SAP, RBC-SAP, XRD patterns of (c) GO, CNT, RBC (d) pure SAP, GO-SAP, CNT-SAP, RBC-SAP, (e) SEM images of GO-SAP, CNT-SAP, RBC-SAP

adsorption and diffusion of water molecules, which is beneficial for enhancing the resin's water absorption capacity. By comparing the SEM images of the three composite SAP, it can be observed that the size and distribution of the pores formed by the addition of different carbon materials vary. GO-SAP and RBC-SAP display relatively larger and more uniformly distributed pores, whereas CNT-SAP exhibits smaller and more densely packed pores. This discrepancy may be attributed to the morphological characteristics of the carbon materials and their dispersion states within the resin matrix.

The influence of carbon materials on the water absorption performance

A single-factor experiment was conducted to optimize the dosage of carbon material. While keeping other conditions constant, the amounts of carbon material added were 0 g, 0.04 g, 0.08 g, 0.12 g, and 0.16 g, respectively. The experimental results are shown in Fig. 3(a). As the amounts of GO, CNT, and RBC increased, the water absorption capacities of the composite SAP exhibited a similar trend: they initially increased and then decreased. The highest water absorption capacities of GO-SAP, CNT-SAP, and RBC-SAP were achieved at GO, CNT, and RBC loadings of 0.08 g, 0.12 g, and 0.12 g, respectively, with values of 251.2 g/g , 200.2 g/g , and 230.2 g/g . This indicates that there exists an "optimal loading" for each of the three carbon materials in terms of enhancing the water absorption performance of the composite SAP.

At low loadings, GO, CNT, and RBC act as physical cross-linking points or reinforcing fillers. The functional groups on their surfaces interact with the resin matrix through hydrogen bonds, ionic bonds, and van der Waals forces, supplementing or strengthening the existing chemical cross-linking network [16]. The hydrophilic groups on these carbon materials increase the overall hydrophilicity of the network, facilitating the attraction of more water molecules. Additionally, the high specific surface areas of these carbon materials provide additional water storage space for the composite SAP. However, when the loading of carbon materials is too high, an excessive number of physical cross-linking points is introduced, leading to an overly dense cross-linking network. This dense network significantly restricts the extension and movement of polymer chains, hindering effective swelling of the network and resulting in a decrease in water absorption capacity. This is the primary reason for the observed trend [17].



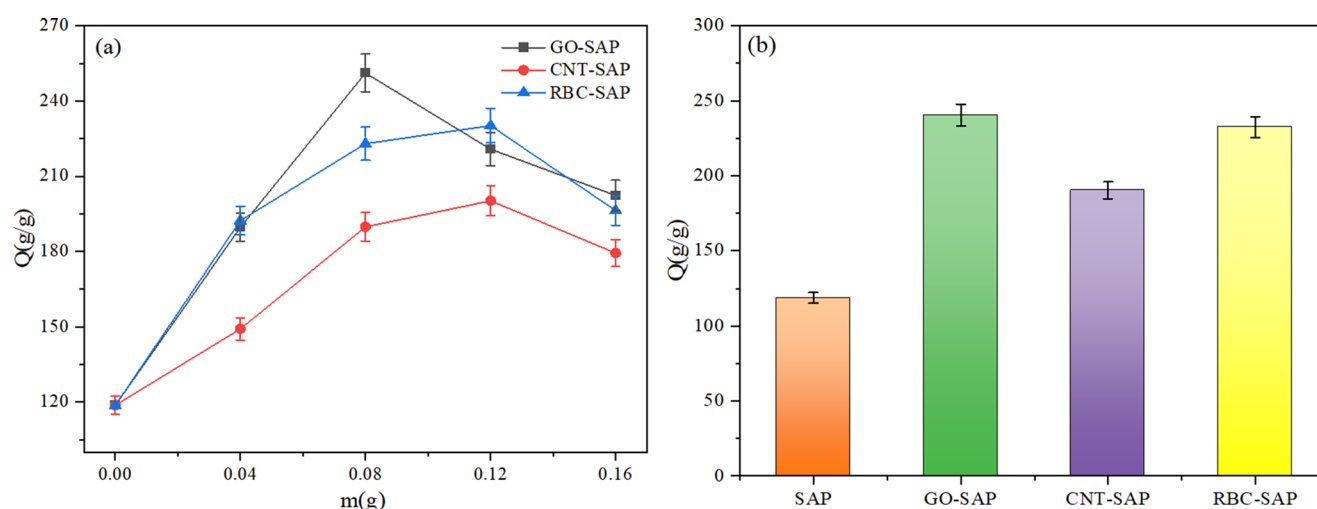


Fig. 3 (a) The influence of carbon material loading on water absorption performance (b) The swelling ratios of pure SAP and composite SAP with a carbon material loading of 0.1 g

For the three composite SAP, the optimal loadings of the carbon materials differ. The inflection point of this “initial increase followed by decrease” trend depends on the filler’s own dispersibility, hydrophilicity/hydrophobicity, size and morphology, as well as its compatibility with the matrix. Among the three, GO-SAP exhibits the highest water absorption capacity at the same loading. This is primarily attributed to a synergistic effect derived from its unique structure and properties. Firstly, the distinctive two-dimensional sheet structure of GO acts as a supporting scaffold within the polymer network, promoting the formation of a more porous yet stable three-dimensional architecture. This provides improved space for water storage and facilitates molecular diffusion [18]. Secondly, the abundant oxygen-containing functional groups, such as hydroxyl and carboxyl groups, on the GO surface not only exhibit high intrinsic hydrophilicity but also form extensive hydrogen bonds with the polymer chains [19]. This significantly strengthens the water-binding capability of the entire network and enhances the driving force for swelling. In contrast, biochar possesses an irregular structure with limited functional groups, while carbon nanotubes are prone to aggregation and inherent hydrophobicity. Consequently, CNT and RBC are less effective than GO in optimizing the network morphology and providing sufficient hydrophilic sites [20, 21].

To systematically compare the effects of different types of carbon materials on the performance of composite SAP, the composite SAP used in subsequent studies in this research were prepared with a carbon nanomaterial loading of 0.1 g. As shown in Fig. 3(b), the water absorption capacities of GO-SAP, CNT-SAP, and RBC-SAP in deionized water were 240.45 g/g, 190.2 g/g, and 232.4 g/g, respectively.

The influence of external factors on the swelling behavior of composite SAP

The impact of pH on the water absorption performance is illustrated in Fig. 4(a). The water absorption capacities of these SAP under different pH conditions exhibit a trend of “initial increase followed by decrease”, maintaining high values within the pH range of 5–10. This phenomenon is fundamentally attributed to the pH-dependent ionization states of key hydrophilic groups (primarily $-\text{COOH}$ and $-\text{SO}_3\text{H}$) on both the resin matrix and the surfaces of the three carbon materials [22]. Under acidic conditions, protonation leads to charge loss and insufficient electrostatic repulsion, causing network contraction. In neutral to weakly alkaline conditions, full deprotonation generates strong electrostatic repulsion, leading to full network expansion. However, under strongly alkaline conditions, high ionic strength induces charge screening and a decrease in osmotic pressure, weakening electrostatic repulsion and causing network contraction [23]. The similarity in oxygen-containing functional groups enriched on the three carbon nanomaterials results in a highly consistent pH-responsive behavior of the composites after their incorporation into the resin matrix. This also indicates that the swelling behavior of SAP is primarily controlled by the charge state and osmotic pressure of the entire cross-linked polymer network [24]. Carbon materials integrate into this network through their surface functional groups, and their behavior is incorporated into the overall network response. Although the swelling of carbon materials themselves is limited, changes in their surface charges significantly affect their interactions with polymer chains (enhancing or weakening repulsive forces) and the overall charge density of the network. The

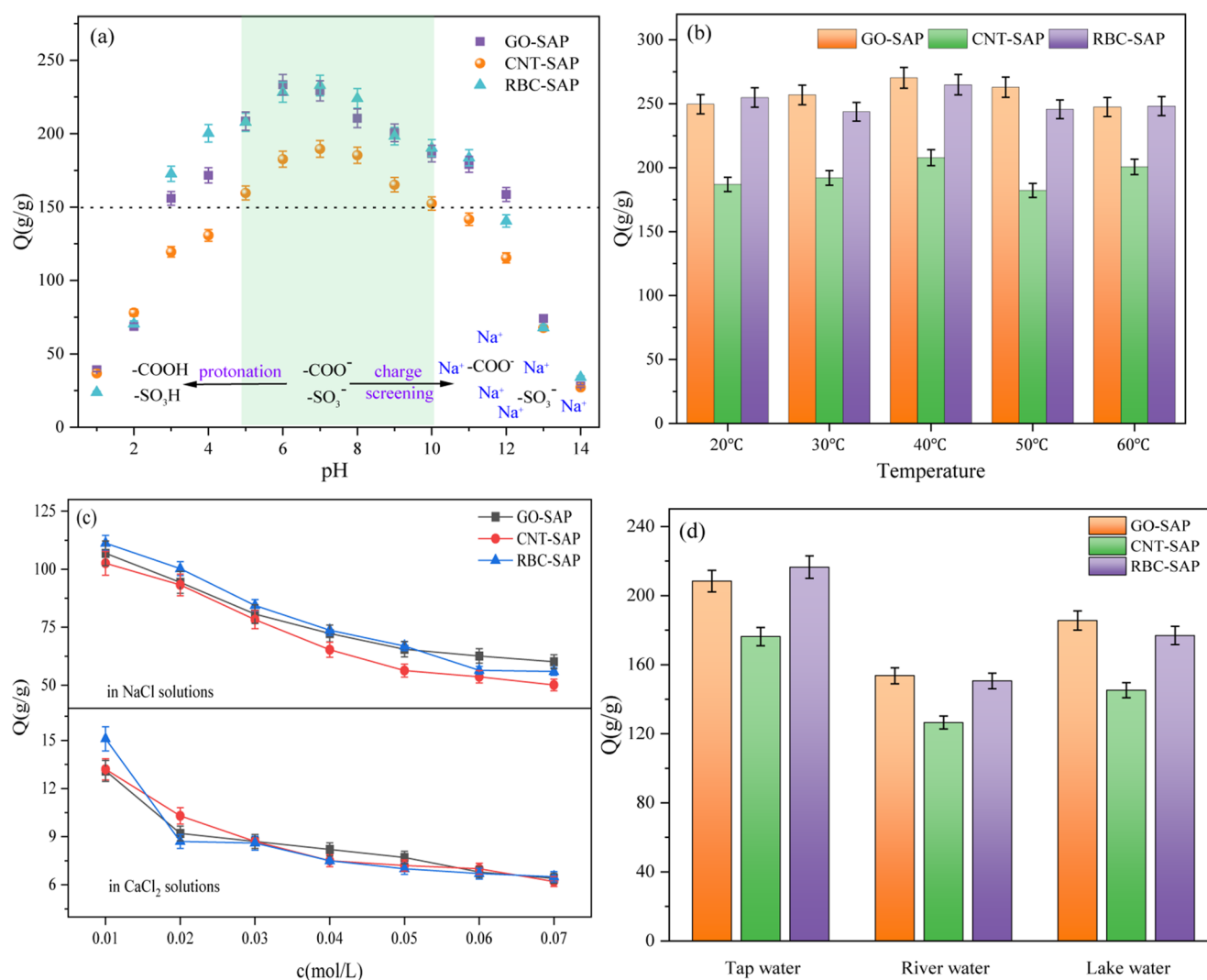


Fig. 4 The effects of (a) pH (b) temperature (c) salts (d) water quality on swelling behavior

experimental data demonstrate that, except under strongly acidic or strongly alkaline conditions, the three composite superabsorbent resins can maintain relatively high swelling ratios, exhibiting good acid-base adaptability.

Figure 4(b) depicts the swelling performance of the three composite SAP at different temperatures. As shown in the figure, the water absorption capacities change little with increasing temperature, indicating good temperature resistance. This suggests that the chemical cross-linking network structure of the resin matrix is stable, and the strength of cross-linking points is not significantly affected by temperature. Additionally, the three carbon materials themselves possess high thermal stability and, when embedded into the network as reinforcing phases and physical cross-linking points, significantly enhance the overall structural rigidity and thermal stability of the composites. This effectively inhibits excessive thermal motion of polymer chains during heating and maintains the integrity of the network structure.

Figure 4(c) presents the water absorption performance of the three composite SAP in NaCl and CaCl_2 solutions of varying concentrations. The figure reveals that, as the concentration of salt solutions increases, the water absorption capacities gradually decrease. Moreover, at the same salt concentration, the water absorption capacity in NaCl solution is significantly higher than that in CaCl_2 solution. The fundamental reason for this phenomenon is that external salt ions inhibit resin swelling by disrupting the core driving forces (osmotic pressure and electrostatic repulsion) that maintain high water absorption [25]. The effect of multivalent cations is more pronounced and complex than that of monovalent cations. High-salt environments significantly reduce the ion concentration difference between the interior of the resin network and the external solution, weakening the osmotic pressure that drives water molecules into the network [26]. External cations ($\text{Na}^+/\text{Ca}^{2+}$) penetrate the network, screening its negative charges and weakening

the electrostatic repulsion that maintains network swelling, leading to network contraction. Compared to Na^+ , divalent Ca^{2+} has a stronger attraction to negative charges and a higher screening efficiency. Ca^{2+} can simultaneously bridge two $-\text{COO}^-$ groups, forming additional non-covalent cross-linking points. This sharply increases the cross-linking density of the resin, greatly limits network expansion, and causes a sudden decrease in water absorption capacity.

Furthermore, we investigated the swelling performance of the three composite SAP in actual water bodies. The results, shown in Fig. 4(d), indicate that, compared to deionized water, the water absorption capacities in three different water bodies decrease to varying degrees. Actual water bodies have higher ionic strengths than deionized water, and natural organic matter or colloidal particles that may be present in river or lake water can adsorb onto the resin surface, hindering water contact with hydrophilic groups or occupying swelling space [27]. This suggests that factors such as ionic strength, natural organic matter, and colloidal particles in actual water bodies significantly impact the swelling performance of composite SAP, necessitating full consideration of these factors in practical applications.

Swelling kinetics

The swelling kinetics of three composite SAP was investigated. As illustrated in Fig. 5(a), all three composite SAP demonstrated excellent initial swelling rates, with their water-absorption capacities rising rapidly during the initial stage of exposure to the aqueous environment. In deionized water, all samples approached the swelling saturation state within 60 min, exhibiting highly efficient water-absorption

capabilities and rapid swelling responsiveness. This is of great significance for practical applications where rapid absorption and retention of large amounts of water are required.

In most cases, the pseudo-first-order kinetic model is only applicable to the initial stage of the swelling process and is not suitable for the entire process. Therefore, the pseudo-second-order kinetic Eq. (3) was employed to explore the swelling processes of the composite SAP [28]. The fitting results, as shown in Fig. 5(b), reveal that the swelling kinetic data are in excellent agreement with the pseudo-second-order kinetic model, with linear correlation coefficients (R^2) all greater than 0.99. This indicates that the water absorption process is primarily controlled by the swelling driving force, and the rate limiting step is dominated by the chemical interactions between the hydrophilic groups on the polymer chains and water molecules, rather than a simple water molecule diffusion process. The high degree of fit of this model further corroborates the previous analysis of the material's structural characteristics [29]. On the one hand, the abundant oxygen-containing functional groups on the surfaces of GO, CNT, and RBC provide additional hydrophilic sites, enhancing the initial binding force with water molecules and accelerating the water-absorption process. On the other hand, the network structures or hydrophilic channels formed by the carbon materials within the polymer matrix synergistically facilitate the penetration and transport of water molecules within the resin. The swelling kinetics results further confirm that the composite SAP not only possess high equilibrium water-absorption capacities but also exhibit rapid response water absorption characteristics, providing crucial kinetic performance support for applications in complex environments.

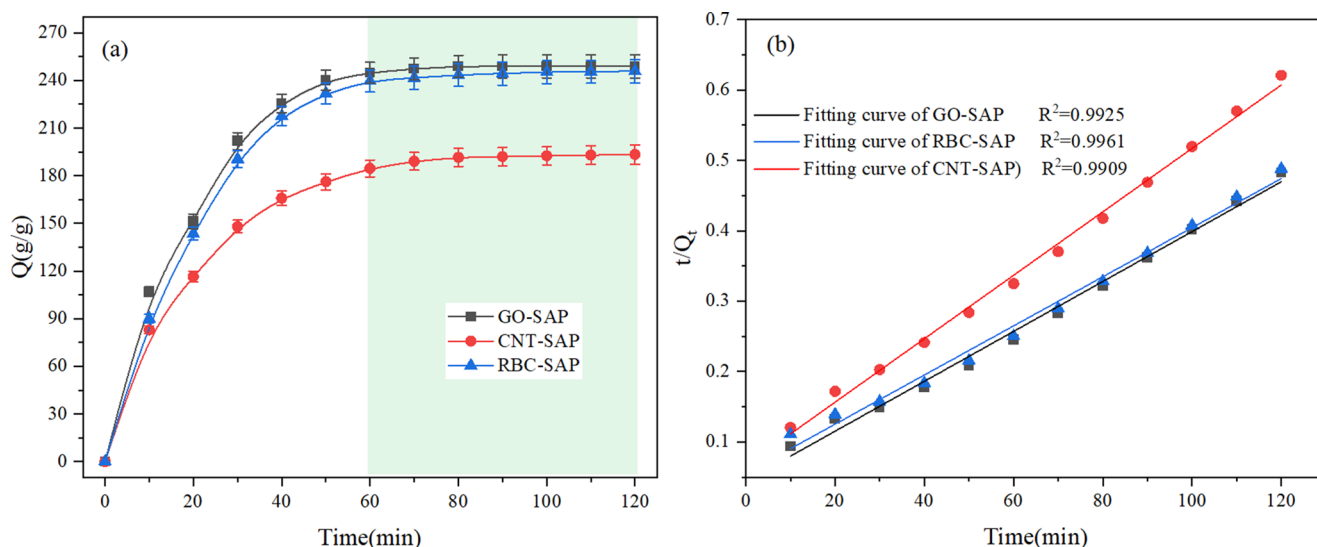


Fig. 5 (a) Swelling rates and (b) pseudo-second-order fitting curves of the composite SAP

$$\frac{t}{Q_t} = \frac{t}{Q_e} + \frac{1}{k} \frac{t}{Q_e^2} \quad (3)$$

Where Q_t is the swelling ratio at time t , g/g; t is the swelling time, min; Q_e is the equilibrium adsorption capacity, g/g; k is the pseudo-second-order swelling rate constant.

Water retention performance

The water retention capacity stands as one of the key performance indicators for SAP's practical applications. In this study, we systematically investigated the water retention characteristics of three composite SAP under static conditions, as well as in thermal and centrifugal force fields.

As illustrated in Fig. 6(a), when placed in a static environment at room temperature, all three materials demonstrated exceptional water-holding stability. It took approximately 10 days for them to lose all the absorbed water, indicating that their network structures can effectively trap free water molecules and inhibit natural evaporation.

In the thermal field, as shown in Fig. 6(b), the rate of water loss accelerated significantly with increasing temperature. In oven environments set at 30 °C, 40 °C, and 50 °C, the time required for the resins to lose all water was 16 h, 7.5 h, and 4.5 h, respectively. This phenomenon aligns with the Arrhenius kinetics, where elevated temperatures speed up the thermal motion of water molecules, thereby weakening the hydrogen bonding between water molecules and the hydrophilic groups of the polymers. However, it is noteworthy that

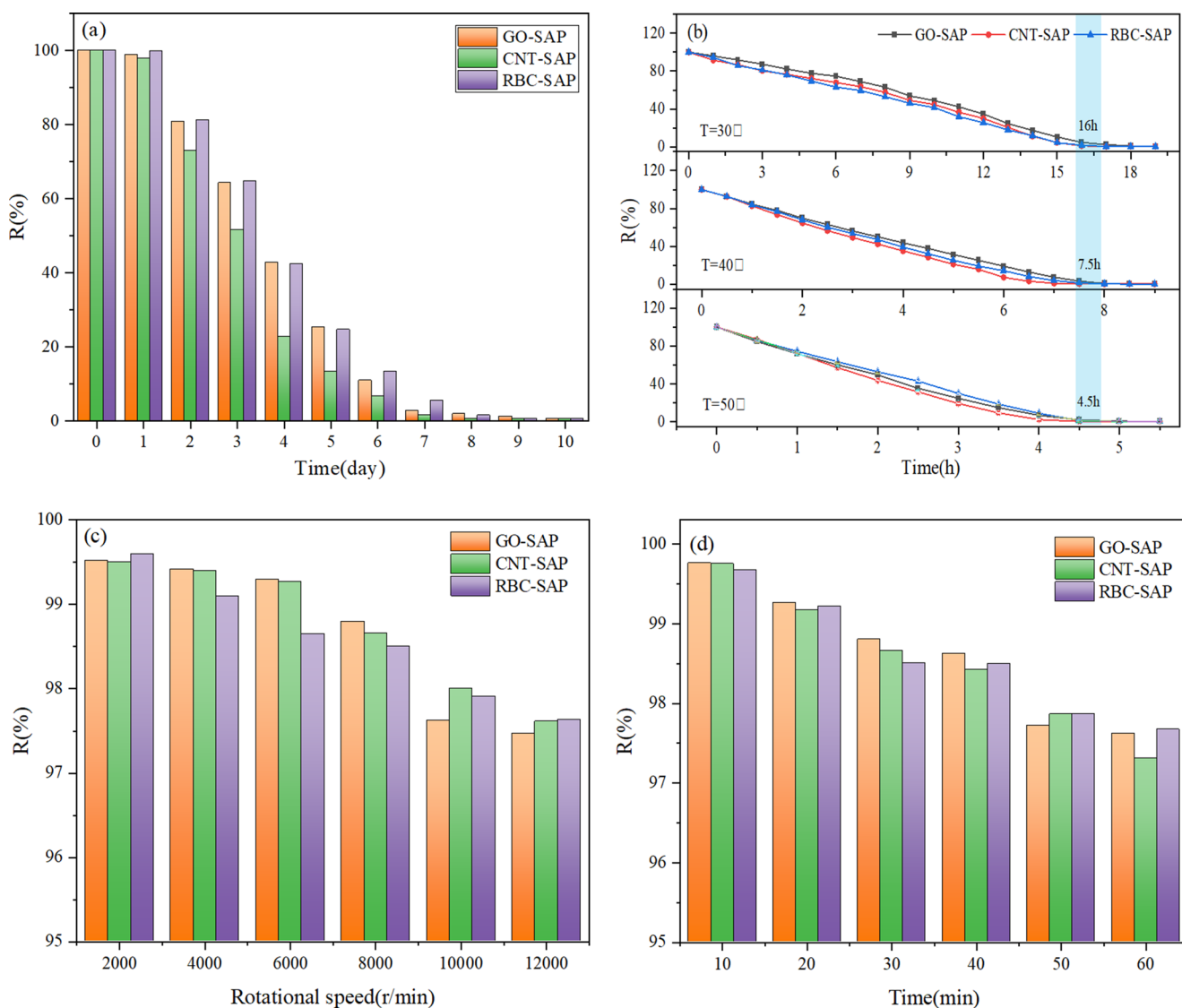


Fig. 6 Water retention performance of SAP under different conditions (a) natural standing (b) forced-air drying at different temperature (c) various rotational speeds for 30 min (d) centrifugation at 8000 rpm for 10–60 min

even at a high temperature of 50 °C, the materials could still maintain their water-retaining capability for several hours, which corroborates the thermal stability of the cross-linked networks revealed in our previous research.

Centrifugal force field tests were further conducted to quantify the materials' resistance to mechanical dehydration. As depicted in Fig. 6(c), under a fixed time condition of 30 min, as the rotational speed increased from 2000 r/min to 12,000 r/min, the water retention rate showed a stepwise decline. Nevertheless, even at the extreme rotational speed of 12,000 r/min, the resins still maintained a high water retention rate of over 97%. Additionally, when the rotational speed was fixed at 8000 r/min, the water retention rate decreased gradually over a period of 10 to 60 min, yet it remained above 97% at the 60-minute mark (Fig. 6d). These two sets of data collectively suggest that while centrifugal force can partially overcome the binding energy between water molecules and hydrophilic groups, leading to the desorption of some free water, the composite SAP possess an extremely high energy barrier for water molecule binding. Even when subjected to a sustained centrifugal force exceeding $10,000\times g$ for up to one hour, they only lost less than 3% of their bound water. This provides compelling evidence for the presence of robust hydration anchoring points within their networks.

Based on comprehensive evaluation across multiple stress fields, the composite SAP demonstrates exceptional structural integrity and hydration stability. The material exhibits prolonged water retention capability under ambient conditions, confirming its stable three-dimensional network architecture. Thermal resistance is evidenced by maintained hydration stability under elevated temperatures, suggesting preserved polymer-water interactions. Most importantly, the material displays remarkable mechanical stability against

intense centrifugal forces, indicating strong binding energetics between water molecules and the polymer matrix. These collective findings substantiate the robust cross-linked network design, highlighting the material's reliable performance under various operational stresses and its potential for practical implementations.

Reusability performance

At the application level, cyclic stability is a critical factor governing the service life and economic viability of materials. To evaluate the practical durability of SAP, six consecutive absorption–drying cycles were performed using three types of composite SAP in two different media: 0.01 mol/L NaCl solution and lake water. As shown in Fig. 7, after six repeated hydration–dehydration cycles, the average water absorption capacity of all three composite SAP remained above 80% of their initial saturated water uptake. These results clearly demonstrate that the as-prepared composite SAP exhibit excellent reusability, structural fatigue resistance, and outstanding cyclic stability. This finding further confirms their long-term reliability in practical applications, providing crucial experimental evidence for the design of high-performance renewable superabsorbent materials.

As shown in Fig. 8, the realization of this stability is primarily attributed to the following synergistic mechanisms. The stable chemical cross-linking points within the resin matrix can effectively resist stress induced damage during the repeated swelling shrinking processes, thereby maintaining the integrity of the network structure. Rigid carbon materials, embedded in the polymer matrix as physical cross-linking points, not only inhibit the irreversible slippage of polymer chains through the steric hindrance effect, but their

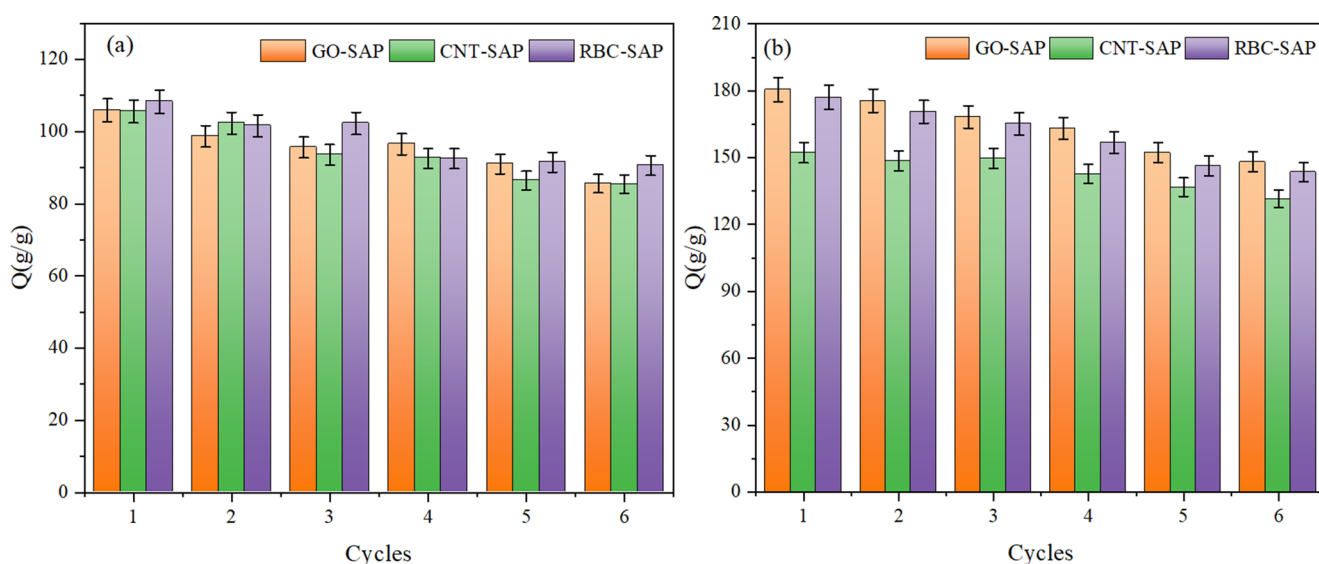


Fig. 7 Repeated water absorption performance of SAP in (a) 0.01 mol/L NaCl solution and (b) lake water

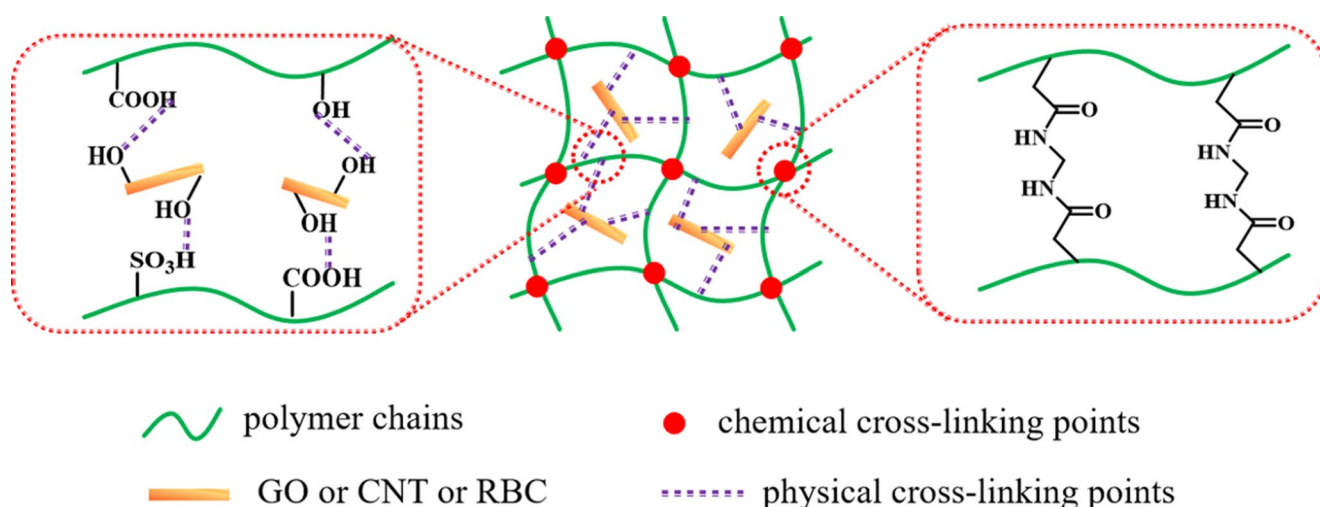


Fig. 8 Schematic diagram of the cross-linked structure of composite SAP

high mechanical strength also disperses the cyclic stress, mitigating the plastic deformation of the network structure. Meanwhile, the abundant hydrophilic groups on the surface of the carbon materials preferentially form hydrogen bonds during the dehydration process, protecting the ionization sites on the polymer backbone from chemical fatigue damage caused by repeated ionization-protonation cycles.

In addition, we compared the reusability performance in lake water of the composite SAP prepared in this study with that of SAP reported in existing literature, and the results are presented in Table 1. The composite SAP we developed demonstrates significant advantages in terms of both the number of recycling cycles and its water-absorption performance over multiple cycles. Consequently, this composite SAP hold greater promise and higher practical value in actual applications, such as agriculture, horticulture, and industrial wastewater treatment. It is expected to offer more efficient and environmentally friendly solutions for these industries.

Table 1 Comparison of the composite SAP and other SAP on reuse performance

SAP	Cycles	Proportion relative to the first full water absorption	Reference
GO-SAP	6	82.0%	This study
CNT-SAP	6	86.4%	This study
RBC-SAP	6	81.1%	This study
FCP-g-PAA/Kaolinite SAP	6	60%	[30]
BP/DAC-g-P(AA-co-IA) SAP	3	80%	[31]
SAS-SBSH SAP	5	64.68%	[32]
HEC-g-P (AA-co-AMPS)/laterite	5(40 °C)	64.3%	[33]
S-SAP	5	<50%	[34]

Mechanism analysis

To elucidate the synergistic enhancement mechanisms of GO, CNT, and RBC in the P(AA-AMPS)/SA, we systematically analyzes their respective roles within the composite system based on the experimental results and previous studies. Owing to their distinct microstructures and surface characteristics, these carbon nanomaterials improve the polymer matrix properties through different pathways: GO, with its two-dimensional layered structure and abundant oxygen-containing functional groups, serves as an efficient physical cross-linker, forming an extensive hydrogen-bonding network with the P(AA-AMPS)/SA chains [35]. This not only enhances the rigidity of the three-dimensional gel network and effectively suppresses excessive swelling and structural disintegration but also provides additional binding sites for water molecules due to its intrinsic hydrophilicity, thereby simultaneously improving the water absorption rate and swelling stability of the material [36]. CNT, utilizing its one-dimensional nanofiber structure, high aspect ratio, and outstanding mechanical modulus, intertwines and interconnects within the polymer matrix to form a continuous three-dimensional supporting framework. This network significantly enhances the toughness, elastic modulus, and fracture resistance of the hydrogel through an effective “fiber-bridging” and stress-dispersion mechanism, ensuring structural integrity even under fully swollen conditions and thus indirectly optimizing water retention and durability [37]. In the case of RBC, its functionality primarily stems from its inherently high specific surface area, well-developed hierarchical porous structure, and surface functional groups. RBC particles act as physical fillers that reinforce the matrix via interfacial interactions, while their internal nanoscale pores directly immobilize a substantial amount of free water through strong capillary adsorption. This water

storage mechanism operates independently of the hydration process of the polymer network, contributing significant additional water capacity to the composite system and directly increasing the equilibrium swelling ratio of the material [38].

Conclusion

Herein, we have successfully fabricated a series of high-performance carbon-composite SAP by incorporating GO, CNT, or RBC into a copolymer network based on AA, AMPS and SA. Among these composites, the graphene oxide modified SAP demonstrates the highest water absorption capacity in deionized water. All synthesized composites maintain stable swelling performance across a broad pH range from 5 to 10 and temperature range from 20 to 50 °C. They also exhibit remarkable water retention capabilities, preserving over 97% water after high-speed centrifugation and maintaining structural integrity for more than 4.5 h at elevated temperature of 50 °C. Furthermore, the materials show excellent reusability, retaining over 80% of their initial swelling capacity after six consecutive absorption-desorption cycles. The swelling kinetics, well-described by a pseudo-second-order model, confirm that the water absorption process is primarily governed by chemical mechanisms. This study establishes a reliable foundation for the development of efficient and sustainable superabsorbent materials, offering promising solutions for addressing water scarcity challenges in sustainable agriculture.

Author contributions Xiangpeng Wang and Yunxiang Zheng performed the experiment, Xiangpeng Wang wrote the manuscript; Chunxiao Zhang performed the data analyses; Chunmao Chen contributed to the conception of the study. All authors reviewed the manuscript.

Funding This work was supported by a Grant from Dongying scientific development fund (DJ2023006) (DJ2023008), the Programs of the Shandong Institute of Petroleum and Chemical Technology (HYKY2023014) (HYKY2023001) and the Carbon Neutrality Research Institute Fund (CNIF20240111).

Data availability All data generated or analyzed during this study are included in this published article.

Code availability Not applicable.

Declarations

Ethics approval Not applicable.

Consent to participate Not applicable.

Consent for publication The paper has not been published elsewhere and that it has not been submitted simultaneously for publication elsewhere. The authors agree to publication in the journal.

Conflict of interest The authors declare there is no conflict of interest regarding the publication of this paper.

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